Experimental Sensitivity Limits and Testability of Electromagnetic Stress Asymmetry in Resonant Cavities

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Abstract

This work presents a quantitative assessment of the experimental testability of electromagnetic stress asymmetry in resonant cavity systems within the framework of Emergent Vacuum Response Theory (EVRT). Using experimentally realistic force sensitivity limits and scaling relations, the accessible parameter space for detecting or constraining effective stress-asymmetry effects is mapped. Current experimental capabilities imply constraints near $\chi_{\text{eff}} \lesssim 10^{-17}$ for representative tabletop resonator parameters, while improved force sensitivity may probe significantly lower regimes. The analysis establishes explicit relationships between measurable force thresholds and theoretical parameter bounds, providing a clear pathway for experimental validation or falsification.

1 Introduction

Previous work has established scaling relations, numerical predictions, and quantitative constraints on electromagnetic stress asymmetry in resonant cavity systems. The present work evaluates the experimental feasibility of detecting such effects by connecting theoretical bounds to measurable force sensitivities.

The central question is not whether an anomalous signal has already been observed, but what level of experimental sensitivity is required to confirm, constrain, or rule out EVRT-like behavior. Framing the problem in terms of sensitivity thresholds and null-result interpretation places the model within a falsifiable experimental structure.

2 Experimental Sensitivity Landscape

Modern precision force measurement systems, including torsion balances and thrust stands, can reach sensitivities in the approximate 1–100 nN range under favorable conditions. These instruments define the practical detection threshold for any EVRT-like effect, directly linking measurable force limits to bounds on the effective response parameter.

Using the reduced scaling relation

$$\chi_{\text{eff}} \sim \frac{F_{\text{noise}} V}{APQ}, \tag{1}$$

where F_{noise} is the experimental force noise floor, V is effective cavity volume, A is the asymmetry parameter, P is input power, and Q is the resonator quality factor, force sensitivity can be translated directly into an upper bound on χ_{eff}.

Figure 1 shows that the constraint improves linearly with force sensitivity. A one-order-of-magnitude improvement in force sensitivity produces a corresponding one-order-of-magnitude improvement in the bound on χ_{eff}.

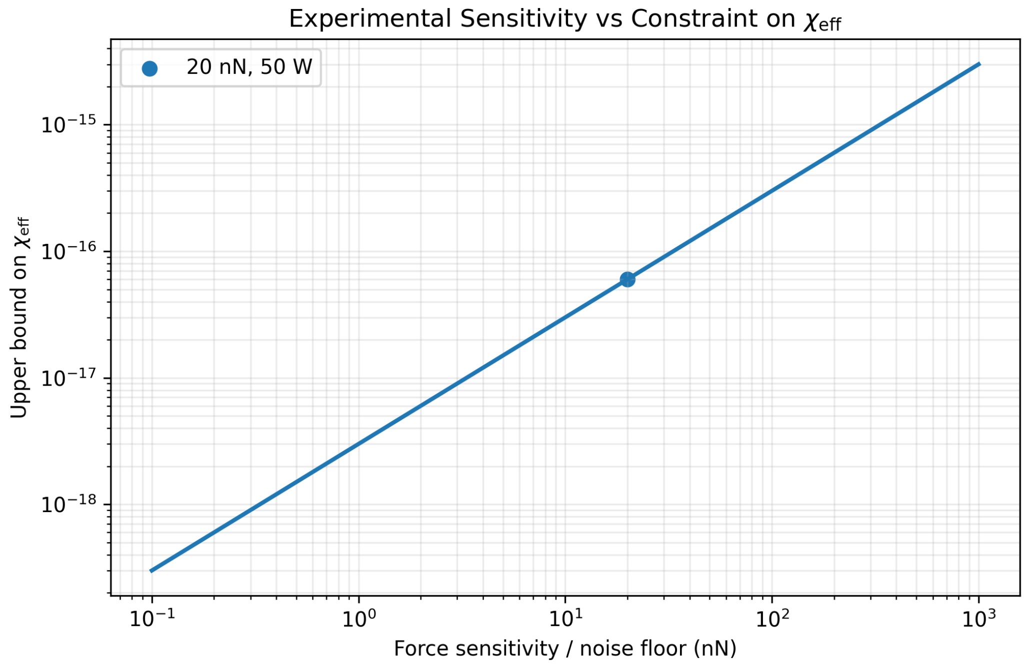

Figure 1: Relationship between experimental force sensitivity and the corresponding upper bound on χ_{eff}, assuming $P = 50$ W, $Q = 10^4$, $V = 1.5 \times 10^{-4}$ m^3, and $A = 0.1$. Improved force sensitivity directly tightens the allowed parameter space.

3 Power Scaling and Detectability

The scaling behavior also shows that increasing input power reduces the minimum detectable value of χ_{eff}. However, practical constraints such as thermal loading, electromagnetic coupling, mechanical drift, and cavity stability impose limits on achievable gains.

Figure 2 illustrates the inverse dependence of the constraint on input power. This relation motivates higher-power testing but also highlights the importance of thermal and electromagnetic controls.

4 Null Result Interpretation

A null result is scientifically meaningful when it is tied to a clearly defined sensitivity threshold. If no force signal is observed above a given threshold, then the corresponding value of χ_{eff} is excluded. This establishes a direct falsification criterion for EVRT-like effects:

$$\chi_{\text{eff}} < \frac{F_{\text{threshold}} V}{APQ}. \tag{2}$$

For example, for $F_{\text{threshold}} = 10$ nN, $P = 50$ W, $Q = 10^4$, $V = 1.5 \times 10^{-4}$ m^3, and $A = 0.1$, the corresponding bound is

$$\chi_{\text{eff}} \lesssim 3 \times 10^{-17}. \tag{3}$$

Thus, an experiment that observes no force signal above 10 nN under these representative conditions would exclude effective response parameters above the 10^{-17} scale.

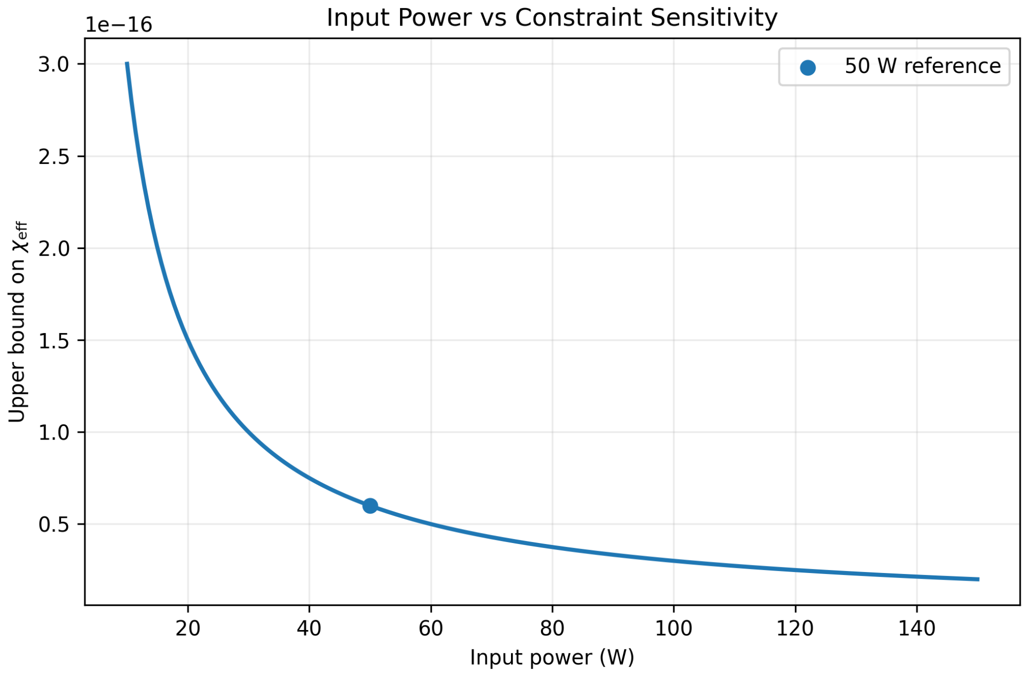

Figure 2: Input power versus achievable constraint sensitivity for fixed $F_{\mathrm{noise}} = 20$ nN, $Q = 10^4$, $V = 1.5 \times 10^{-4}$ m^3, and $A = 0.1$. Higher power improves the bound, but only if systematic artifacts remain controlled.

5 Representative Sensitivity Limits

Sensitivity (nN)	Power (W)	χ_{eff} Limit
20	50	6×10^{-17}
10	50	3×10^{-17}
1	100	3×10^{-18}

Table 1: Representative experimental sensitivity levels and corresponding constraints on χ_{eff}, assuming $Q = 10^4$, $V = 1.5 \times 10^{-4}$ m^3, and $A = 0.1$.

Table 1 summarizes how experimental improvements translate into tighter bounds. The key result is that sensitivity gains and power increases act multiplicatively in reducing the allowed parameter space.

6 Discussion

These results place the EVRT framework within the domain of experimentally testable physics. Existing measurement capabilities already constrain large regions of parameter space, and incremental improvements in sensitivity can further tighten these bounds.

This analysis does not assert the existence of electromagnetic stress-asymmetry forces. Instead, it provides a method for interpreting positive and null experimental outcomes. A positive result would require reproducible scaling with power, resonance, asymmetry, and orientation controls. A null result at sufficient sensitivity would progressively exclude larger regions of the effective-response parameter space.

The most important implication is that EVRT-like effects are not merely conceptual: they can be mapped onto explicit experimental thresholds. This strengthens the connection between theoretical modeling, constraint analysis, and future validation efforts.

7 Conclusion

This work establishes a direct mapping between experimental sensitivity and theoretical parameter bounds for electromagnetic stress asymmetry in resonant cavity systems. By defining explicit exclusion regions for χ_{eff}, it provides a robust framework for systematic validation or falsification of EVRT-like models.

Future experimental advances in force sensitivity, thermal control, cavity stability, and systematic-error rejection will further refine these constraints and clarify the viability of electromagnetic stress-asymmetry effects under nonequilibrium resonant conditions.